

# Activity 2.1: ANOVA & Statistical Model for a CRD

## Traffic Signals

### Note

Submit via Gradescope, don't forget to assign pages to the questions in the outline!  
Note: while you are using the .qmd or Word template the links won't work – make sure to also reference the online instructions!!

*I worked on this activity with:* \_\_\_\_\_

A traffic engineering study was conducted to investigate traffic delay at urban intersections. Three types of traffic signals were studied: pretimed, semi-actuated, and fully actuated. Fifteen intersections were completely randomly assigned to one of the three types of signals. The response variable was the average stopped time per vehicle (seconds) at each intersection.

The data for this study is provided in [traffic\\_data.csv](#).

Download the data here or on Canvas and save it in your nicely organized “activities” or “data” folder!

This is what the data looks like:

| id | traffic_signal | avg_stop_time_secs |
|----|----------------|--------------------|
| 1  | Pretimed       | 36.6               |
| 2  | Pretimed       | 37.1               |
| 3  | Semi-actuated  | 18.7               |
| 4  | Fully actuated | 15.0               |
| 5  | Fully actuated | 18.9               |
| 6  | Fully actuated | 15.2               |
| 7  | Pretimed       | 39.2               |
| 8  | Semi-actuated  | 17.5               |
| 9  | Pretimed       | 30.4               |
| 10 | Semi-actuated  | 25.7               |

| id | traffic_signal | avg_stop_time_secs |
|----|----------------|--------------------|
| 11 | Fully actuated | 10.4               |
| 12 | Fully actuated | 10.5               |
| 13 | Semi-actuated  | 22.0               |
| 14 | Pretimed       | 34.1               |
| 15 | Semi-actuated  | 20.6               |

### a. Study Structure

Describe the following components of the study.

- Treatment structure
- Design structure
- Response variable

### b. Complete the skeleton ANOVA table below

Make sure to use context of the data!

|                          |                   |
|--------------------------|-------------------|
| Source of Variation (SV) | DF: Total = _____ |
|                          |                   |
|                          |                   |
|                          |                   |
|                          |                   |

### c. ANOVA Table

Use the data provided to calculate the following (you can use R):

$$\bar{y}_{..} = \underline{\hspace{2cm}}$$

$$\bar{y}_{1.} = \underline{\hspace{2cm}}$$

$$\bar{y}_{2.} = \underline{\hspace{2cm}}$$

$$\bar{y}_{3.} = \underline{\hspace{2cm}}$$

for  $i = 1$  (Pretimed),  $i = 2$  (Semi-actuated),  $i = 3$  (Fully actuated)

- Show (by hand, using a calculator) how the SSE is calculated:

Then, use your SSE calculation and the provided information to fill out the ANOVA table below. *You can check your answers by running the ANOVA in R, but know where the values come from.*

| Source | DF | SS      | MS | F | Prob > F |
|--------|----|---------|----|---|----------|
|        |    |         |    |   | <0.0001  |
| Total  | 14 | 1340.45 |    |   |          |

### d. Statistical Model

Write the *effects model* (sum to zero) for this study, make sure to provide the index values, assumptions, and clearly define each term in context.

**e. Using R, obtain the parameter/coefficient estimates for the following:**



**Reminder for R**

When using R, you need to change the default settings to use “sum to zero”:

```
options(contrasts = c("contr.sum", "contr.poly"))  
mod <- lm(...)
```

Check your factor order with:

```
levels(data$variable_name)
```

- $\hat{\mu} =$  \_\_\_\_\_
- $\hat{\tau}_1 =$  \_\_\_\_\_
- $\hat{\tau}_2 =$  \_\_\_\_\_
- $\hat{\tau}_3 =$  \_\_\_\_\_

for  $i = 1$  (Pretimed),  $i = 2$  (Semi-actuated),  $i = 3$  (Fully actuated).

**f. Interpret Parameter Estimates**

In context of the data, provide an interpretation for  $\hat{\tau}_1 = 12.02$ . It may be helpful to think about what  $\bar{y}_1$  was from (c).

**g. Provide a conclusion**

In context of the data, write out your conclusion from the overall ANOVA F-test from (c). Make sure to provide all evidence – (F = , df = , p = ).